HR Employee Attrition Analysis

Exploratory Data Analysis and Visualization Project

Tools: Python (Pandas, Matplotlib, Seaborn) | Platform: Google Colab

Total Employees	Employees Left	Attrition Rate	Avg Monthly Income
1,470	237	16.12%	6,503

1. Project Overview

Employee attrition (employees leaving the company) is costly: it brings recruitment expense, training effort and loss of experience. This project analyses an IBM HR dataset to understand **who is leaving, where, and why**, so that HR can take targeted retention action.

Objectives

- Measure the overall attrition level in the company.
- Understand the profile of employees (age, income, job role, department).
- Identify the factors most strongly linked to attrition.
- Recommend practical retention actions based on evidence.

2. Dataset Description

Item	Detail
Source file	IBM_HR_Attrition_Enhanced.csv
Rows (employees)	1,470
Columns	39 (demographics, job, pay, satisfaction, tenure)
Missing values	None
Target variable	Attrition (Yes / No)
Derived columns	AgeGroup, TenureGroup, MonthlyIncomeBracket, OverallSatisfactionIndex

Data preparation

- Dropped columns that add no information: EmployeeCount, StandardHours, Over18 (same value for every row) and EmployeeNumber (only an ID).
- Created a 0/1 flag from Attrition so that the average of the flag equals the attrition rate for any group.
- Because the data is imbalanced (1,233 stayed vs 237 left), groups are compared using **percentage rates**, not raw counts.

3. Methodology

The analysis moves from simple to complex, and every chart is followed by an interpretation.

Level	Variables	Question answered	Charts used
Univariate	1	What does each column look like?	Countplot, histogram
Bivariate	2	Which factors relate to attrition?	Bar chart, boxplot
Multivariate	3+	Who exactly is at risk?	Heatmap, scatterplot

4. Univariate Analysis

Each variable is studied on its own to understand its distribution.

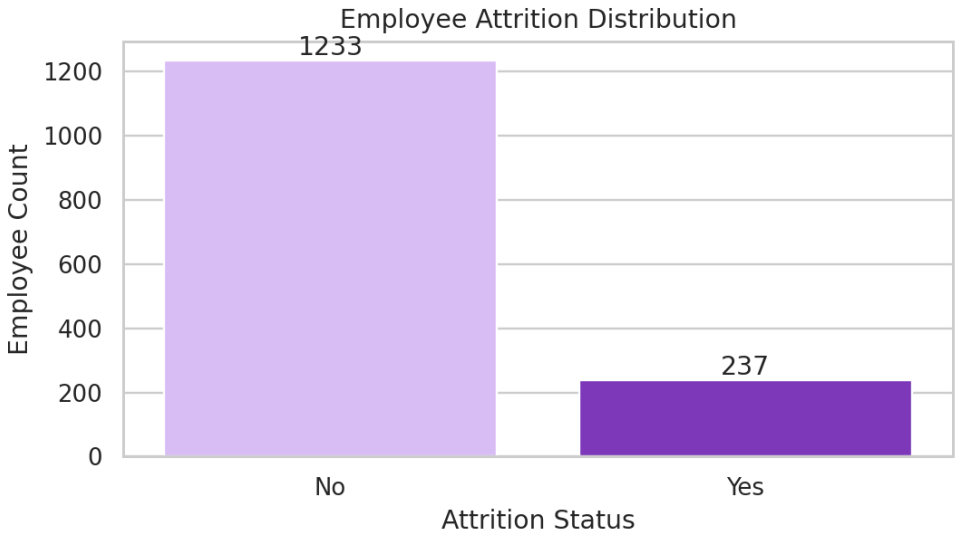


Chart 1: Attrition count

Interpretation: Out of 1,470 employees, 237 left, an attrition rate of **16.12%**. The data is imbalanced, so later comparisons use rates. A 16% turnover is significant and carries real replacement cost.

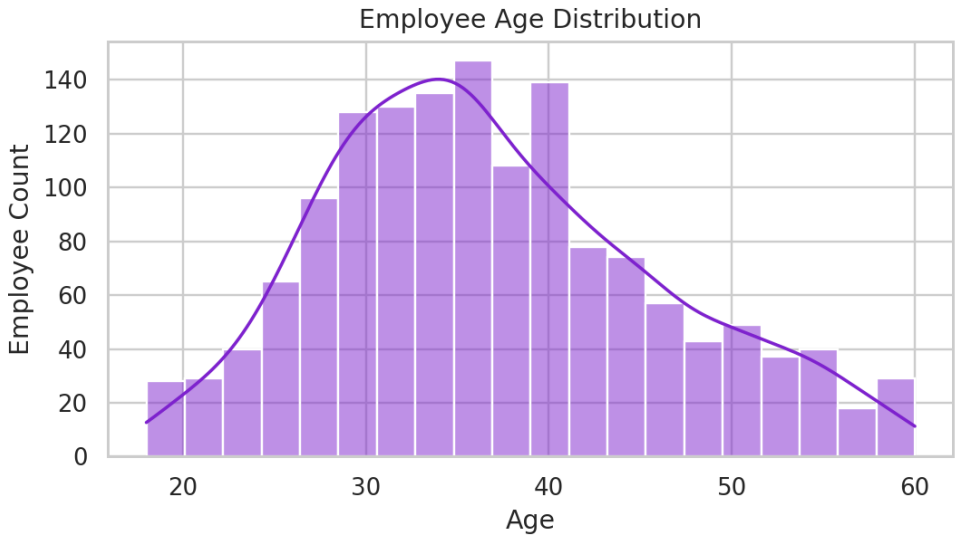


Chart 2: Age distribution

Interpretation: Ages range from 18 to 60, with most employees between **30 and 40** (average about 37). The distribution is only slightly right-skewed, so the workforce is mainly young to mid-career professionals.

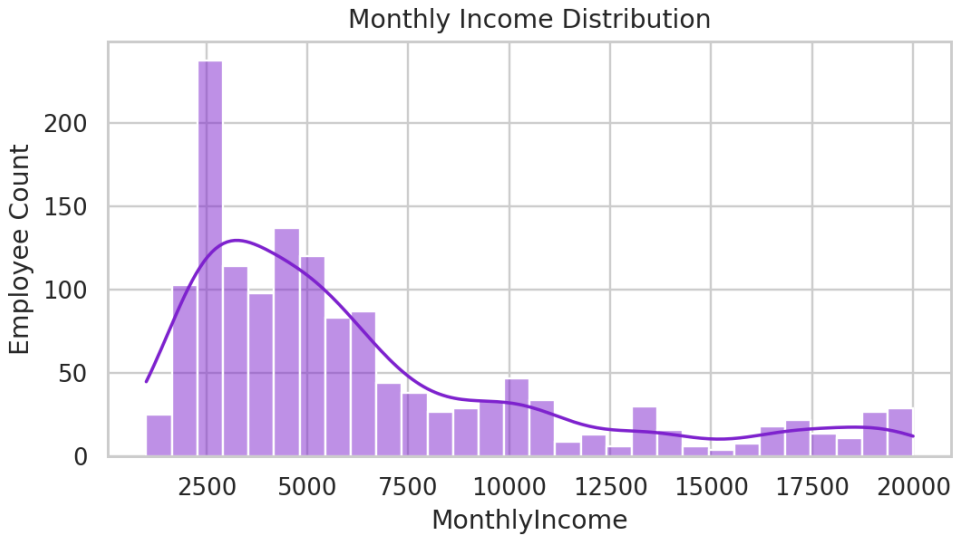


Chart 3: Monthly income distribution

Interpretation: Income is **right-skewed**: most employees earn low to medium salaries (around 2,500 to 7,500), while a small group earns very high pay up to 20,000. A few high salaries pull the average (6,503) above the typical value, so the median is a fairer measure. The company is junior-heavy.

5. Bivariate Analysis

Each factor is compared with Attrition to find relationships.

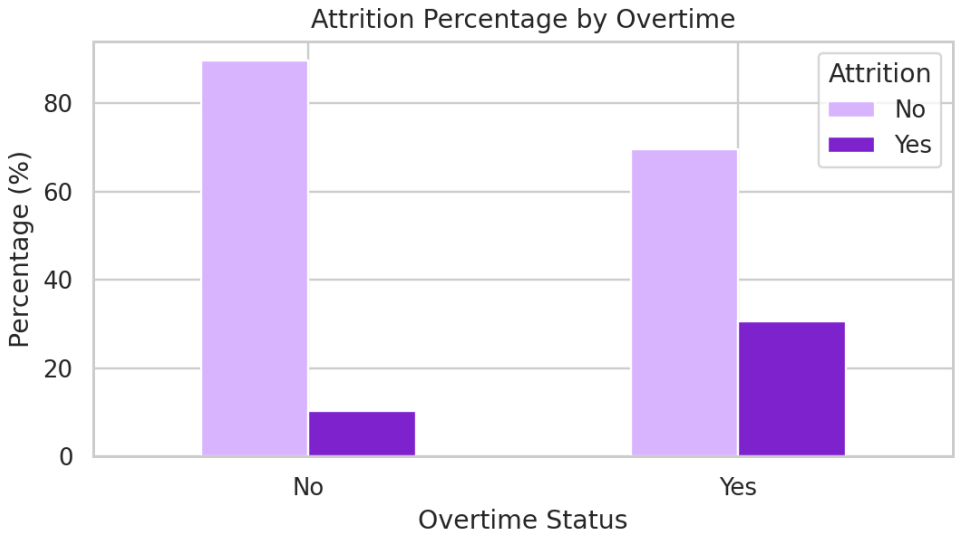


Chart 4: Attrition percentage by overtime

Interpretation: About **30.5%** of overtime workers left compared to **10.4%** of those without overtime, roughly **3 times higher**. Overtime is the strongest single driver of attrition. **Action:** cap overtime hours and review workload.

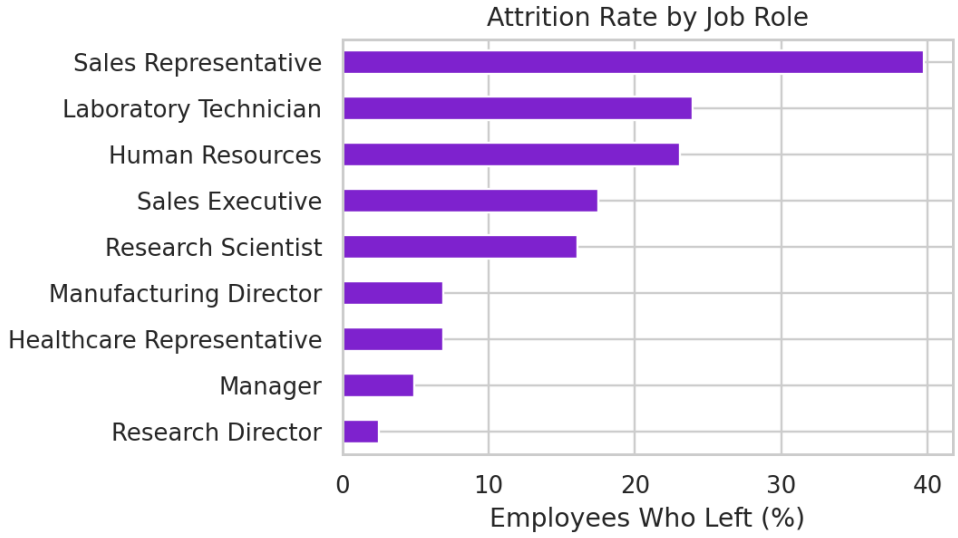


Chart 5: Attrition rate by job role

Interpretation: **Sales Representatives** have the highest attrition (about 40%), followed by Laboratory Technicians (about 24%) and Human Resources (about 23%). Senior roles are very stable: Managers (about 5%) and Research Directors (about 2.5%). Retention effort should focus on junior sales and technical staff.

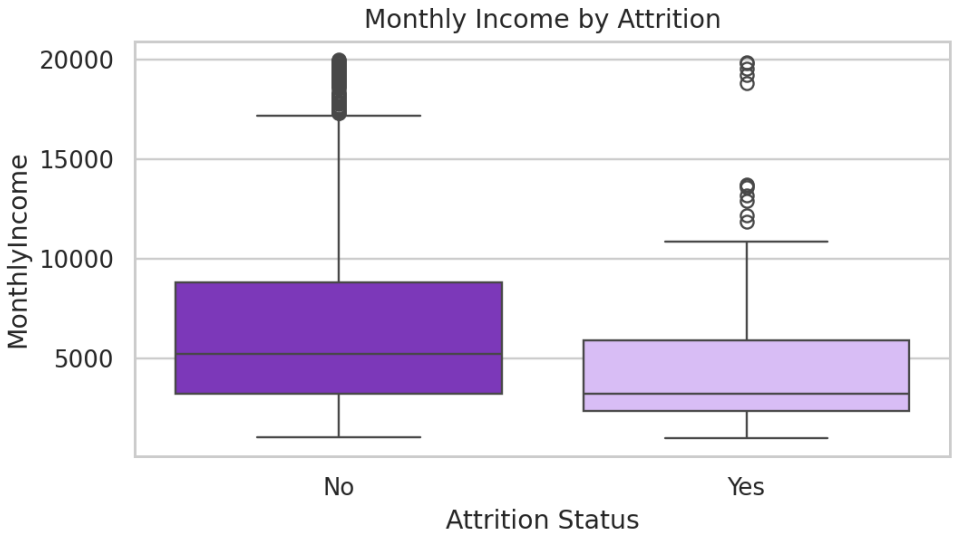


Chart 6: Monthly income by attrition

Interpretation: The median income of employees who left is about **3,200**, versus about **5,200** for those who stayed. Leavers are paid noticeably less, so low pay is a clear exit signal.

6. Multivariate Analysis

Three or more variables are combined to find the deeper story.

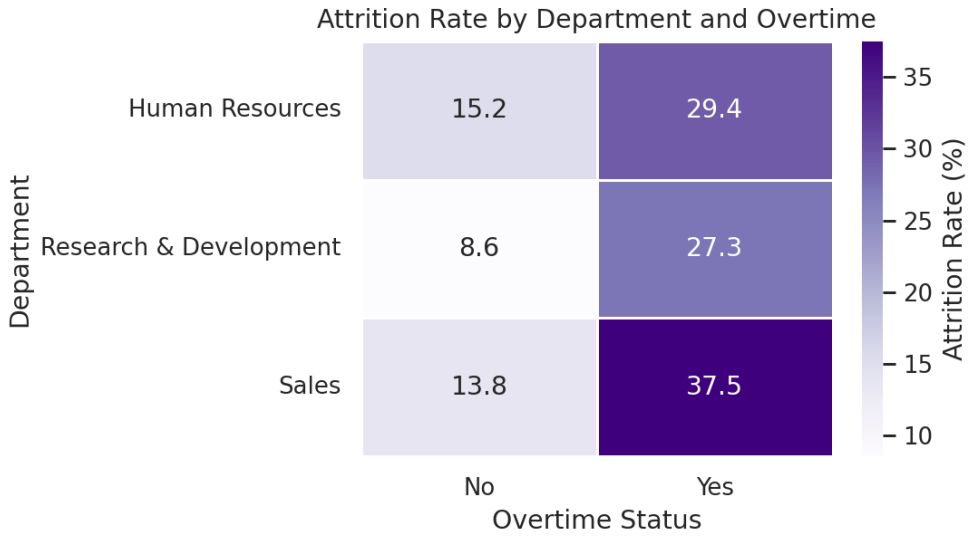


Chart 7: Attrition by department and overtime

Interpretation: Overtime raises attrition in **every** department. **Sales with overtime is the worst at 37.5%.** R&D is the most stable without overtime (8.6%) but jumps to 27.3% with overtime, which shows overtime is the real cause, not the department itself. **Action:** reduce overtime in Sales first.

Department	No Overtime (%)	Overtime (%)
Human Resources	15.2	29.4
Research & Development	8.6	27.3
Sales	13.8	37.5

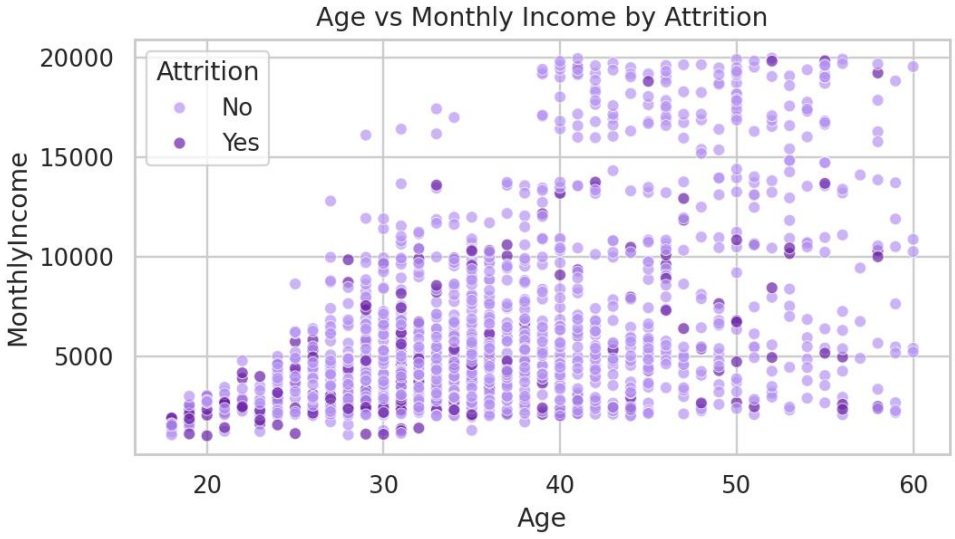


Chart 8: Age vs monthly income by attrition

Interpretation: Income rises with age. Employees who left (dark dots) cluster in the **bottom-left corner**: young (about 18 to 35) and earning below about 5,000. Few leave at higher income or older age. Attrition is concentrated among early-career, low-paid staff.

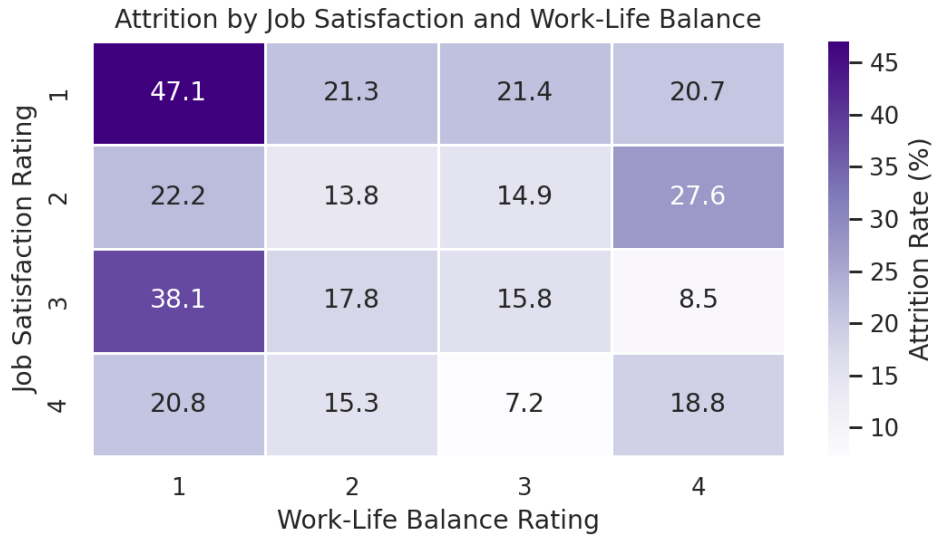


Chart 9: Attrition by job satisfaction and work-life balance

Interpretation: The highest attrition (**47.1%**) occurs where job satisfaction is 1 and work-life balance is 1, the lowest rating on both. Low work-life balance combined with low satisfaction is a strong warning sign. The pattern is not perfectly smooth in the middle ratings, so this chart is best used to spot extreme cases.

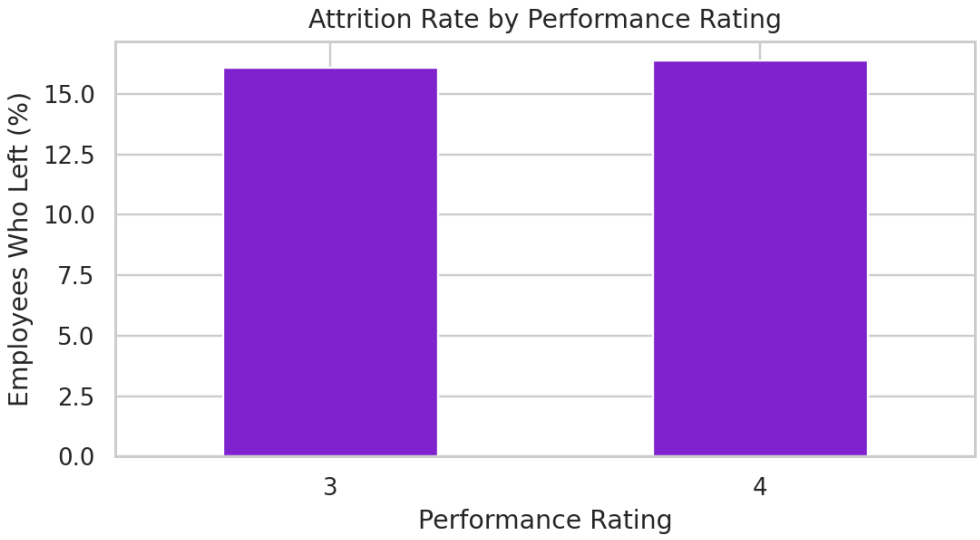
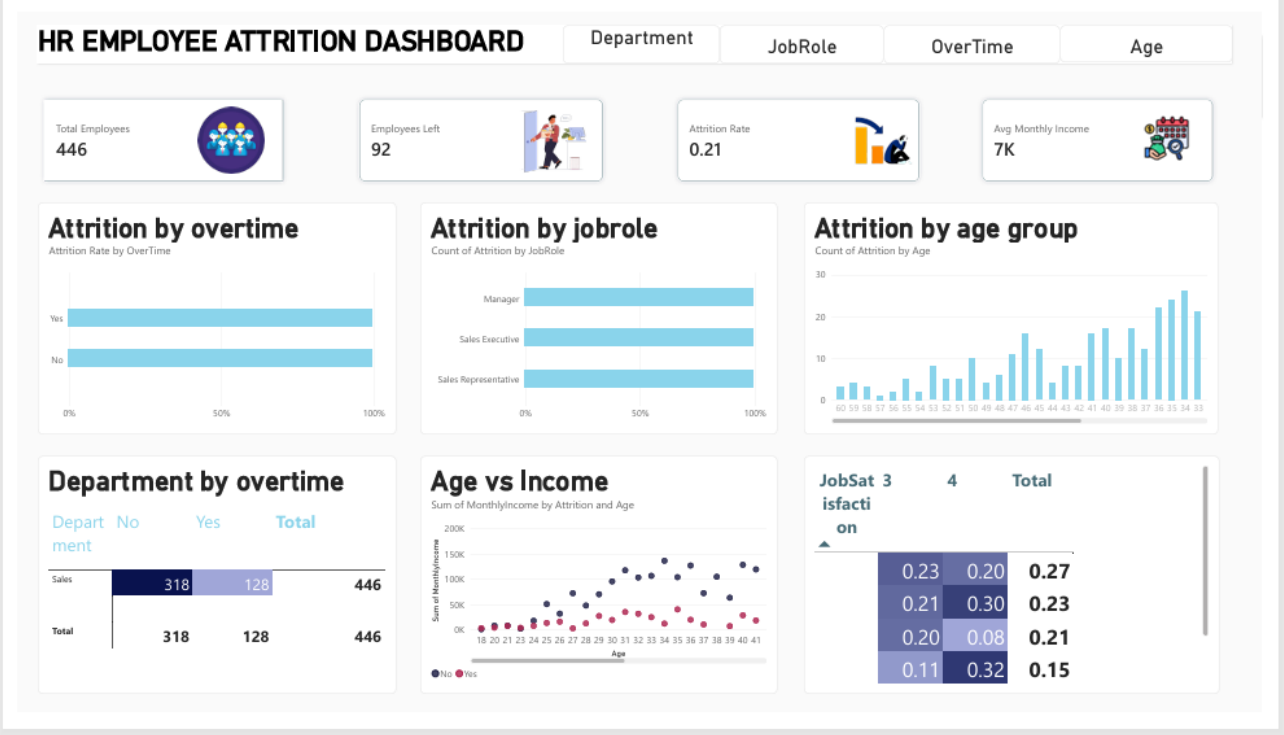


Chart 10: Attrition rate by performance rating

Interpretation: Attrition is almost the same for rating 3 (16.1%) and rating 4 (16.4%). **Performance rating does not explain attrition.** Ratings 3 and 4 are the only values present, and high performers leave as often as others, so the company loses talent at the same rate.

Final Dashoard View



7. Key Business Insights

- **Overall attrition is 16.12%** (237 of 1,470 employees).
- **Overtime is the top driver:** 30.5% attrition with overtime versus 10.4% without.
- **Who leaves:** young employees, low-income employees, and those in their first years at the company.
- **Where they leave from:** Sales Representatives (about 40%), Laboratory Technicians and HR. Sales with overtime reaches 37.5%.
- **Pay matters:** median income of leavers (about 3,200) is far below that of stayers (about 5,200).
- **Work-life balance and satisfaction:** employees rating both at the lowest level show the highest attrition (47.1%).
- **Performance is not a factor:** high and average performers leave at the same rate.

8. Recommendations

Problem found	Recommended action
High overtime attrition	Set overtime limits, hire additional staff in Sales, monitor overtime hours monthly
Young and low-paid staff leaving	Review entry-level pay, introduce career growth plans and mentoring
Sales Representatives (about 40%)	Improve incentives, workload and early-career support in Sales
Low satisfaction and work-life balance	Run satisfaction surveys, offer flexible working and wellbeing programs
Talent lost regardless of performance	Add recognition and promotion paths for high performers

9. Conclusion

The analysis shows that attrition at this company is not random. It is concentrated among **young, low-paid employees working overtime, especially in Sales**. Univariate analysis described the workforce, bivariate analysis found the key drivers (overtime, pay, job role), and multivariate analysis showed exactly which combinations carry the highest risk. Acting on overtime and early-career pay offers the largest opportunity to reduce turnover.

10. Limitations

- The analysis shows association, not proof of cause.
- The data is a single snapshot with no time dimension.
- Differences between groups are not tested statistically in this stage.